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SECOND REVIEW UNDER THE STAFF-MONITORED PROGRAM—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the staffs of the International Monetary Fund (IMF) and the International Development Association (IDA)

Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*High*
Overall risk of debt distress	*High*
Granularity in the risk rating	*Sustainable*
Application of judgement	*No*

Comoros external and overall debt are sustainable but remain at high risk of debt distress.[1] *Breaches of the external debt sustainability thresholds are of limited magnitude while debt service metrics are projected to gradually improve over the forecast horizon. Additional considerations that mitigate the mechanical risk signals include the authorities' successful completion of the IMF Staff-Monitored Program (SMP), plans for an economic reform program supported by the prospective IMF's Extended Credit Facility (ECF), and the consistently strong performance of remittances, which supports external sustainability. The baseline in this DSA reflects higher oil and food prices, the appreciation of the dollar, and a less active policy-adjustment scenario compared to the last DSA (reflecting a pre-ECF baseline). Three of four external debt burden indicators breach their respective thresholds under this baseline. The incorporation in domestic debt of statutory advances, SDR on-lending to the government and higher level of publicly guaranteed domestic debt of State-Owned Enterprises (SOEs), result in deterioration of domestic debt dynamics under the*

[1] Comoros' debt carrying capacity is assessed as medium, given a Composite Indicator of 2.855 based on October 2022 WEO projections and the 2021 Country Policy and Institutional Assessment (CPIA) rating.

baseline. This assessment is contingent on: (i) progress on domestic resource mobilization and gradual but sustained fiscal consolidation over the medium term, (ii) no additional non-concessional borrowing, and (iii) enhanced financial oversight of SOEs. The DSA suggests that Comoros is susceptible to export and exchange rate depreciation shocks; natural disaster and contingent liabilities shocks would also present downside risks to the debt outlook.

PUBLIC DEBT COVERAGE

1. Public debt coverage has been extended to include all financing from the central bank to the Treasury. The coverage of external debt includes the entire public sector, namely external debt of the central government, the central bank borrowing on behalf of the government, and government-guaranteed debt of SOEs. Subnational government entities cannot take up external debt on their own, and SOEs cannot access the external debt market without a government guarantee. Domestic debt includes central government, SOE's guaranteed debt from domestic banks, and on-lending from the central bank including the IMF SDR allocation and statutory advances.

Text Table 1. Comoros: Public Debt Coverage

Subsectors of the public sector	Sub-sectors covered
1 Central government	X
2 State and local government	
3 Other elements in the general government	
4 o/w: Social security fund	
5 o/w: Extra budgetary funds (EBFs)	
6 Guarantees (to other entities in the public and private sector, including to SOEs)	X
7 Central bank (borrowed on behalf of the government)	X
8 Non-guaranteed SOE debt	

Sources: Comorian authorities, and IMF staff

2. Contingent liabilities have built up significantly and the likelihood of their materialization is growing. The magnitude of the contingent liabilities shock is revised up to 12.5 percent of GDP, from 8.8 previously, to reflect: (i) preliminary outcomes of an ongoing audit pointing to a higher-than-expected level of domestic arrears[2] and (ii) difficulties encountered by the SOEs involved in importing food and oil in rolling over their trade loans following the deterioration of their cash flows due to higher commodity prices.

Text Table 2. Coverage of the Contingent Liabilities' Stress Test

1	The country's coverage of public debt	The central government, central bank, government-guaranteed debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	4.5	Reflects unaudited estimates of domestic arrears.
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	3	SOEs growing debt
4	PPP	35 percent of PPP stock	0	
5	Financial market (the default value of 5 percent of GDP is the minimur	5 percent of GDP	5	
Total (2+3+4+5) (in percent of GDP)			12.5	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

Sources: Comorian authorities, and IMF staff

[2] These domestic arrears represent unpaid bills for goods and services and are included as contingent liabilities as the figures are still preliminary. They include arrears from the treasury to the pension system. Preliminary assessment from the ongoing audit is that the accumulation of domestic arrears includes considerable cross debts between the state and SOEs as well as among SOEs, reflecting PFM weaknesses rather than government insolvency. The government plans to elaborate a domestic arrears clearance strategy upon the completion of the audit exercise.

BACKGROUND

A. Recent Debt Developments

3. Public external debt has grown while domestic public debt rose significantly albeit still modest. External public debt reached 28 percent of GDP in 2022 and is expected to rise further to 31.5 percent of GDP by end-2023, up from 20.3 percent of GDP prior to the pandemic. Domestic public debt jumped from 2.9 in 2020 to 5.1 percent of GDP in 2022, reflecting the BCC's on-lending of the SDR allocation a higher take-up of commercial bank debt by the SOEs involved in food and energy supply.

4. Negotiations are ongoing to finalize loans for the El Maarouf Hospital. The authorities are seeking three concessional loans (amounting to USD 70 million), which are included in this baseline.

5. Debt Structure. At end-2021, disbursed debt owed to official bilateral creditors accounted for 63 percent of public external debt, debt to multilateral creditors for 16 percent, and debt to regional lenders for 17 percent. The share of non-concessional debt stood at 14 percent of external debt at end-2021 and is expected to grow in the next few years as the non-concessional loan for the construction of the Galawa Hotel is disbursed.

6. With the expiration of the DSSI and tightening liquidity constraints, the authorities have accumulated new external arrears[3] Between end-December 2021 and end-December 2022, external arrears increased from USD 6 million to USD 7.2 million, of which USD 5.2 million in external arrears (72 percent of the total) are owed to BADEA. The authorities plan to repay shortly arrears owed to the Kuwait Fund (USD 0.12 million) and the OPEC Fund for International Development (USD 0.35 million). Discussions are still ongoing regarding the pre-existing stock of arrears owed to Exim Bank India for a project loan in dispute (USD 1.17 million). External arrears also accumulated vis-à-vis Bpifrance Assurance Export (USD 0.56 million), for which the authorities reported that they are awaiting a response to their request for debt cancellation from the French Treasury.

B. Macroeconomic Forecasts

7. Compared to the last DSA, the macroeconomic assumptions have been adjusted to reflect the impact of the recent shocks and the greater uncertainty in the global outlook going forward (Text Table 4)**.** The previous DSA, prepared in the context of the SMP request, built on an active policy scenario and a more favorable long-term outlook. In contrast, the current DSA is based on a less active policy scenario, reflecting the pre-ECF baseline. Long-run growth has also been revised down from the previous DSA to reflect the effects of consecutive shocks in recent years, delayed reforms as short-term policies prioritized addressing the shocks, and the significantly greater uncertainty in the global outlook.

- Real GDP growth for 2022 has been revised down to 2.4 percent from 3.8 percent before Russia's invasion of Ukraine. Growth is projected to rise to 3.0 percent in 2023 and then to an

[3] Projections of debt service payments to clear the arrears are included in macroeconomic assumptions underlying this exercise. The arrears represent < 1 of GDP and hence do not automatically trigger an in-debt distress rating.

average of 4.1 percent in 2024-28, driven by large investment projects, including hospital expansion, the construction of a hotel complex, and the refurbishment of roads, while remittances are expected to decline from their earlier record levels during the COVID-19 pandemic amid lower growth in the eurozone, albeit remaining strong. In the long run, the boosting effect of those investment projects will dissipate, and growth will come back to an average of 3.8 percent.

Text Table 3. Comoros: Decomposition of Public Debt 2019-21[1]

	2019	2020	2021	2019	2020	2021
		Million USD			% GDP	
Total	280.7	322.7	388.1	22.9	26.1	29.8
External	241.3	282.6	316.9	20.3	23.2	24.7
Multilateral creditors[2]	87.9	104.0	116.5	7.4	8.5	9.1
IMF	19.0	28.7	27.6	1.6	2.4	2.1
World Bank	12.3	17.6	22.0	1.0	1.4	1.7
ADB/AfDB/IADB	2.5	2.5	7.1	0.2	0.2	0.5
Other Multilaterals	54.1	55.2	59.9	4.6	4.5	4.7
o/w: BADEA	28.3	28.0	28.0	2.4	2.3	2.2
o/w: TDB	12.0	13.2	18.3	1.0	1.1	1.4
o/w: IsDB	7.6	7.5	7.5	0.6	0.6	0.6
Bilateral Creditors	153.4	178.6	200.3	12.9	14.7	15.6
Paris Club	2.1	2.1	2.1	0.2	0.2	0.2
o/w: France	2.1	2.1	2.1	0.2	0.2	0.2
Non-Paris Club	151.3	176.5	198.2	12.7	14.5	15.4
o/w: China	66.2	91.7	113.2	5.6	7.5	8.8
o/w: Saudi Arabia	33.7	33.7	33.7	2.8	2.8	2.6
o/w: India	26.4	26.4	26.4	2.2	2.2	2.1
o/w: Kuwait	25.0	24.8	25.0	2.1	2.0	1.9
Bonds	0.0	0.0	0.0	0.0	0.0	0.0
Commercial creditors	0.0	0.0	0.0	0.0	0.0	0.0
Domestic	32.0	36.8	59.1	2.6	2.9	5.1
Held by residents, total[3]	32.0	36.8	59.1	2.7	3.0	4.6
Loans	32.0	36.8	59.1	2.7	3.0	4.6

1/As reported by Country authorities according to their classification of creditors, including by official and commercial.
2/ "Multilateral creditors" are simply institutions with more than one official shareholder and may not necessarily align with creditor classification under other IMF policies (e.g. Lending Into Arrears)
3/ Inludes central bank on lending related to the SDR allocation

Source: Country authorities and staff estimates and projections

- Average inflation is estimated at 12 percent in 2022 percent following higher international prices of oil and food aggravated by shortages linked to shipment difficulties. It will fall to around 8.2 percent in 2023 due to the base effect, the end of shipment difficulties and to the ongoing impact from the tightening of the monetary policy. After 2023, inflation will stabilize at around 2-2½ percent reflecting the peg of the exchange rate to the Euro.

- Fiscal sector: SOE's losses contributed to lower fiscal revenue in 2022 but lower COVID-19-related spending helped narrow the domestic primary deficit from 3.4 percent of GDP in 2021 to 1.7 percent of GDP in 2022. This will be followed by a widening of the primary deficit to 2.7 percent of GDP reflecting higher investment spending during 2023-24, with the primary deficit thereafter averaging at 1.7 percent of GDP per year. Projections include a fiscal financing gap starting in 2023 that mainly reflects insufficient funding for postal bank recapitalization, increased debt service obligations and lower revenues from key state-owned enterprises that need to recoup their earlier losses.

- The external current account deficit is estimated to have widened to 4.5 percent of GDP in 2022 and is expected to widen further in 2023 to 7.3 percent before gradually declining to 4.5 percent. Exports of goods are projected to recover to pre-pandemic levels by 2026 and improve further after that while tourism is already recording a stronger performance compared to pre-COVID-19 levels. Imports jumped to a high level amid the war in Ukraine and will be spurred by substantial public investment in 2023-24. Remittance inflows are expected to remain strong despite the correction from the COVID-19 levels, given the expected continued importance of the Comorian diaspora. [4] With sustained import growth and increased interest payments from previously contracted non-concessional debt, the current account is projected to weaken over the next few years before settling at around 4½ percent of GDP over the long term. Despite initially comfortable foreign exchange reserves coverage, the fiscal financing gap translates into an external financing gap of the same size as the government faces tight domestic and external borrowing constraints.

8. This DSA assumes that the authorities will adopt a prudent borrowing strategy. Staff assumes that disbursement from already contracted loans will continue over the coming years in line with the authorities' plans.

- **Domestic financing:** The drawdown of government's deposit at the central bank during 2022-23 is projected at 2.7 percent of GDP. The other domestic financing in this DSA captures SOEs' short term domestic debt.

- **Foreign financing:** Staff assumes that as yet undisbursed parts of already contracted loans will be disbursed over the coming years in line with the authorities' plans. IDA financing, including through investment projects, is projected at USD 28.7 million per year on average over the next four years. The baseline includes around USD 50 million from AfDB and USD 17 million from Saudi Arabia over the next four years. Staff further assumes that a remaining fiscal gap will be filled with borrowing on concessional terms. This assumption reflects Comoros' commitment to avoid non-concessional borrowing going forward. It is also underpinned by the implementation of performance policy actions (PPAs) under the sustainable development financing policy (SDFP) in 2022-2023 which aimed at improving debt management with a debt management law which was promulgated (in 2023), streamlining the borrowing process, reinforcing the role of the ministry of finance in the borrowing

[4] During 2017-21, the average annual growth rate of remittances was 15 percent.

process, and establishing a committee to advise the minister of finance on the approval of loans and guarantees (in 2022). The average external loan financing from 2022-2032 is USD 62 million per year (including the fiscal financing gap) as compared to USD 59 million from the previous DSA. The average grant element of new borrowing is 54.5 percent, higher than the last DSA when it stood at 43.4 percent.

9. Realism tools do not flag issues (Figures 3 and 4). The 3-year fiscal adjustment is smaller than in most historical Fund-supported programs. The projected GDP growth in 2023 is compatible with a range of plausible fiscal multipliers. The projected contribution of government investment over the medium term is in line with historical experience. Changes between the sum of debt dynamics contributions during 2016-2021 against 2022-2026 reflect the impact of a higher projected GDP growth and current account normalization as the country rebounds from several shocks during 2019-21, the appreciation of the dollars over the next five years, and a higher primary deficit in 2023-24 reflecting higher investment spending.

Text Table 4. Comoros: Assumptions for DSA

	2020	2021 Prel.	2022 Proj.	2023 Proj.	2024 Proj.	2025 Proj.	2026 Proj.	2027 Proj.	2028 Proj.	Long Term[1]
GDP Growth Rate (percent)										
Previous DSA	-0.5	1.6	3.8	3.7	3.9	4.3	4.4	4.2	4.2	4.2
Current DSA	-0.2	2.1	2.4	3.0	3.6	4.1	4.3	4.3	4.2	3.8
Inflation, average (percent)										
Previous DSA	0.8	-1.0	1.2	1.4	1.6	1.7	2.0	2.0	2.0	2.0
Current DSA	0.8	0.0	12.0	8.2	1.6	2.3	2.1	2.0	2.0	2.0
Government Revenues (percent of GDP)										
Previous DSA	9.3	8.9	9.2	9.5	9.8	10.1	10.4	10.7	11.0	13.2
Current DSA	9.2	10.3	9.6	9.5	9.7	10.0	10.2	10.4	10.6	12.1
Overall Fiscal Balance (percent of GDP)										
Previous DSA	-1.9	-3.5	-3.8	-2.1	-0.1	-0.1	-0.2	-0.4	-0.5	-1.1
Current DSA	-1.4	-3.4	-1.7	-2.7	-2.7	-1.4	-1.4	-1.4	-1.4	-1.8
FDI (percent of GDP)										
Previous DSA	0.0	0.3	0.3	0.4	0.6	0.7	0.9	1.0	1.1	1.4
Current DSA	0.3	0.3	0.4	1.5	1.1	0.6	0.6	0.6	0.6	0.6
Current Account Deficit (Percent of GDP)										
Previous DSA	-1.7	-4.1	-7.6	-8.6	-6.5	-6.8	-6.9	-7.2	-7.3	-7.5
Current DSA	-1.9	0.3	-4.5	-7.3	-6.4	-6.1	-6.4	-5.4	-4.7	-4.6
Export of Goods and Services (annual percentage growth)										
Previous DSA	-38.6	22.8	30.3	5.1	6.4	7.1	7.8	7.1	7.3	7.6
Current DSA	-48.3	62.4	19.8	9.2	7.2	8.8	9.0	9.3	9.8	6.8
Import of Goods and Services (annual percentage growth)										
Previous DSA	-3.7	7.8	18.2	4.3	-2.1	7.2	7.3	7.3	6.9	6.6
Current DSA	-2.5	4.6	17.3	18.5	3.7	1.7	6.3	4.7	6.9	6.7
Investment (percent of GDP)										
Previous DSA	15.6	16.2	17.3	16.9	15.4	15.6	15.9	16.1	16.3	17.9
Current DSA	15.6	15.7	16.7	16.9	17.2	17.3	17.5	17.6	17.7	18.3
Net Current transfers (percent of GDP)										
Previous DSA	18.7	16.4	14.7	13.7	13.5	13.5	13.5	13.4	13.5	13.4
Current DSA	20.2	20.8	18.1	17.7	17.1	15.5	14.7	14.7	14.8	15.0

1 Covers the period 2029-42 for current DSA, and 2029-41 for the previous DSA.

Source: Authorities data and IMF staff estimates

C. Country Classification and Determination of Stress Test Scenarios

10. Comoros is assessed to have medium debt carrying capacity, unchanged from the 2021 DSA (Text Table 5), with a Composite Indicator (CI) of 2.855 based on October 2022 WEO projections and the 2021 Country Policy and Institutional Assessment (CPIA) rating. As a result, this DSA continues to use the same external debt burden thresholds and total public benchmarks for countries with medium debt carrying capacity (Text Table 5). The relevant thresholds indicating high risk are 40 percent for the PV of debt-to-GDP ratio, 180 percent for the PV of debt-to-exports ratio, 15 percent for the debt service-to-exports ratio, and 18 percent for the debt service-to revenue ratio. These thresholds are applicable to public and publicly guaranteed external debt.

11. Standardized stress tests use default settings. In addition**,** Comoros qualifies for the natural disaster scenario given that it is exposed and vulnerable to natural disaster shocks, such as cyclones and eruptions of the Karthala volcano. The tailored stress test assumes a one-off natural disaster shock of 10 percent of GDP to external PPG debt-GDP ratio in the second year of the projection period.

Text Table 5. Comoros: Calculation and Contribution of the CI Score Components

Debt Carrying Capacity	Medium

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Medium	Medium 2.85	Medium 2.89	Medium 2.89

Applicable thresholds

External debt burden thresholds	
PV of debt in % of	
Exports	180
GDP	40
Debt service in % of	
Exports	15
Revenue	18

Total public debt benchmark	
PV of total public debt in percent of GDP	55

Source: CI score calculations based on LIF DSF.

DEBT SUSTAINABILITY

A. External Debt Sustainability

12. Three out of four external debt burden indicators breach their respective thresholds under the macroeconomic assumptions underlying this exercise (Figure 1).

- The PV of external debt-to-GDP ratio is still below the threshold but increased compared to the previous exercise.
- The breach of the PV of debt-to-export ratio persists during 2025-27 before declining below the threshold.
- The debt service-to-exports ratio breaches its thresholds during 2025-27 and the debt service-to-revenue ratio during 2025-28. Compared to the previous DSA, the breaches are less pronounced because delays in project execution during 2021-22 have smoothed loan disbursements and, as a consequence, associated debt services.

13. The stress tests suggest that export shock and exchange rate depreciation have the largest negative impact on the debt trajectory. The export shock would cause threshold breaches for three of four indicators while the depreciation shock while the depreciation shock would cause breach to the debt service to revenue ratio. The results indicate that a natural disaster shock may increase the PV of external

debt-to-GDP ratio from 18 percent to 31 percent in the medium term, and debt service to revenue ratio from 8 percent to peak at 21 percent in 2025. The contingent liabilities shock may have a similar impact on PV of external debt-to-GDP ratio and debt service to revenue ratio.

B. Total Public Debt Sustainability

14. Overall public debt is projected to rise under the passive policy scenario but remain well below its indicative benchmark under both the macroeconomic assumptions underlying this exercise and the stress tests (Figure 2). The incorporation in domestic debt of statutory advances, SDR on-lending to the government and higher level of publicly guaranteed domestic debt of SOEs, result in deterioration of domestic debt dynamics, relative to the last DSA, under a less active policy scenario currently assumed in the baseline. The stress tests suggest that export and growth shocks have the largest negative impact on the debt trajectory. The growth shock may increase the debt service to revenue ratio from 7 percent to peak at 63 percent in 2029. The export shock may have similar impact on PV of debt-to-GDP ratio and PV of debt-to-revenue ratio.

C. Risk Assessment and Vulnerabilities

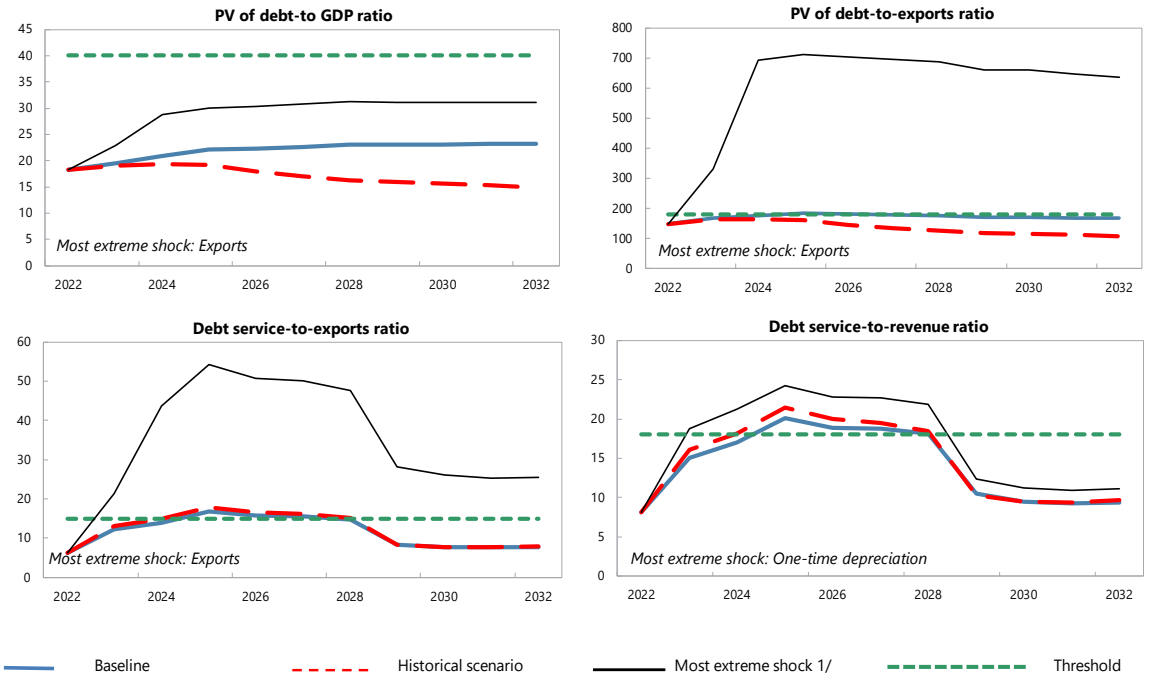
15. Comoros external and overall debt are sustainable but remains at a high risk of external and overall debt distress. The external debt and overall debt continue to be assessed as sustainable, given the limited magnitude of the breaches of the external debt sustainability thresholds and the projected improvement of debt service *metrics over the forecast horizon. Additional considerations that mitigate the mechanical risk signals in support of the analysis include the authorities' plans for an economic reform program to be supported by the prospective IMF's Extended Credit Facility, and the consistently strong performance of remittances, which supports external sustainability.

16. Mitigating risks to public debt requires several policy actions: Building on the successful completion of the SMP, the authorities have requested an ECF program that will help mitigate risks to public debt via (i) making faster progress on domestic resource mobilization and ensuring gradual but sustained fiscal consolidation over the medium term, (ii) avoiding additional non-concessional borrowing, (iii) enhancing financial oversight of SOEs, and (iv) implementing reforms to boost growth.

AUTHORITIES' VIEWS

17. The authorities broadly shared Fund staff's assessment, emphasizing that reducing debt vulnerabilities is a key priority for their development agenda. They emphasized the significant impact of recent successive shocks on the macroeconomic picture and agreed that fiscal reforms will be needed to gradually reduce risks over the medium term. They reiterated their commitment to engaging with staff in the context of a follow-on program from the SMP to push forward planned fiscal and other structural reforms.

Figure 1. Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2022-2032



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	Yes	
Natural disaster	Yes	No
Commodity price 2/	n.a.	n.a.
Market financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

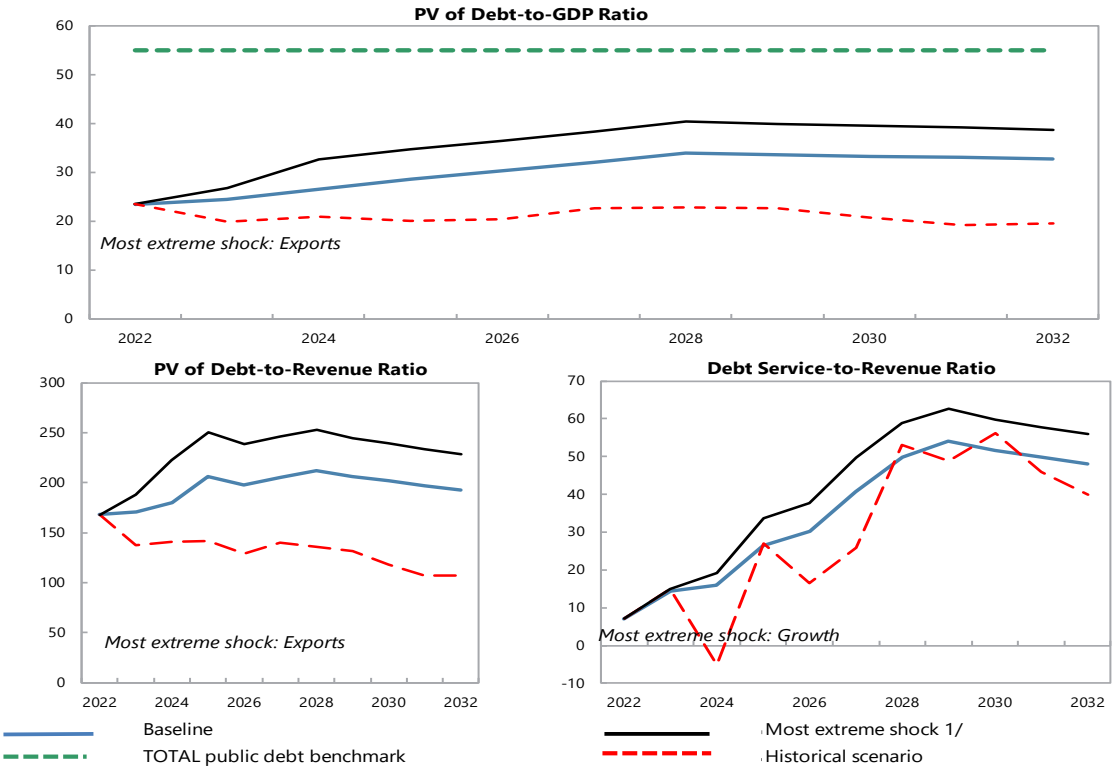
Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.0%	1.0%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	37	37
Avg. grace period	9	9

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in or before 2032. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Indicators of Public Debt Under Alternative Scenarios, 2022-2032



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	49%	49%
Domestic medium and long-term	0%	0%
Domestic short-term	51%	51%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.0%	1.0%
Avg. maturity (incl. grace period)	37	37
Avg. grace period	9	9
Domestic MLT debt		
Avg. real interest rate on new borrowing	0.0%	0.0%
Avg. maturity (incl. grace period)	1	1
Avg. grace period	0	0
Domestic short-term debt		
Avg. real interest rate	0.5%	0.5%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

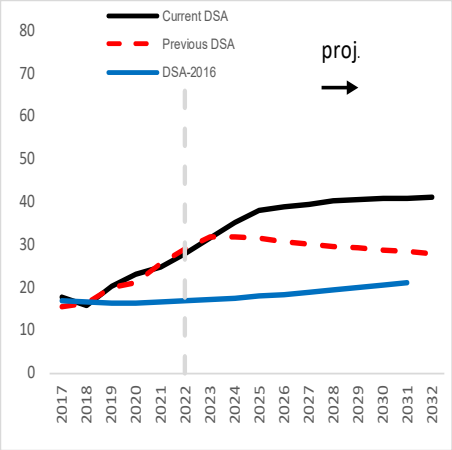
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2032. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

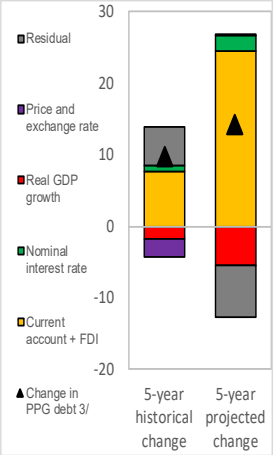
Figure 3. Drivers of Debt Dynamics - Baseline Scenario

External Debt

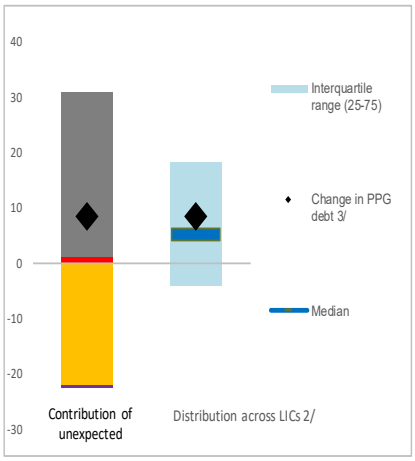
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

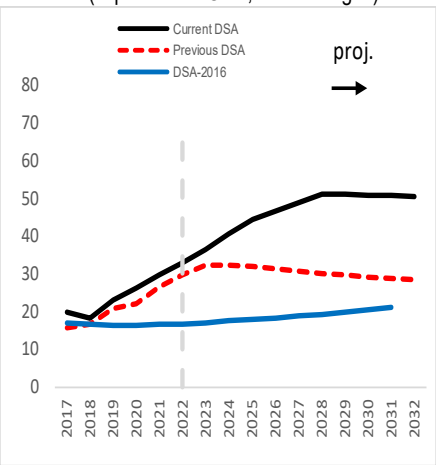


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

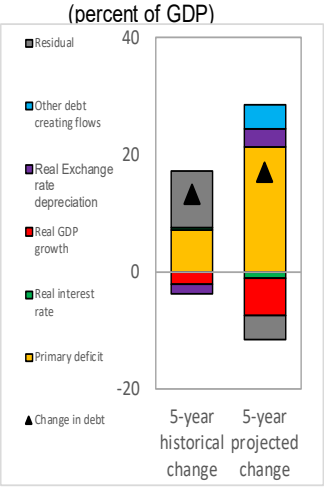


Public debt

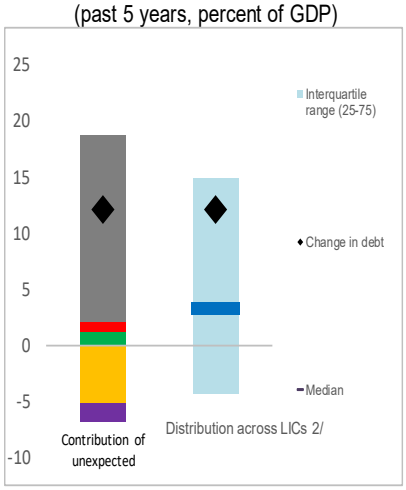
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



Sources: Country authorities; and staff estimates and projections.

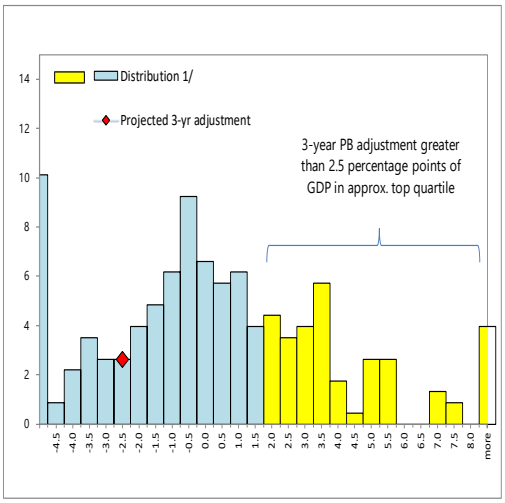
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

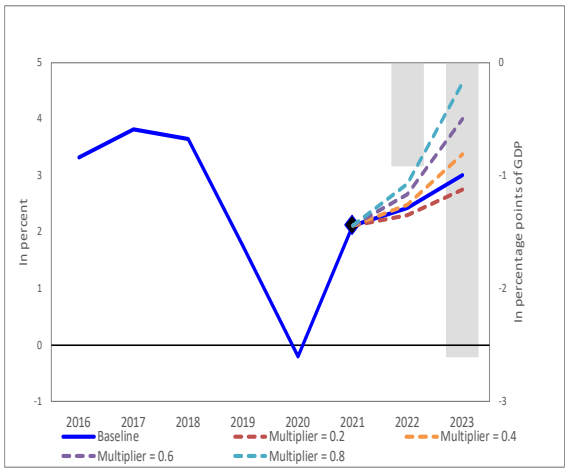
Figure 4. Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



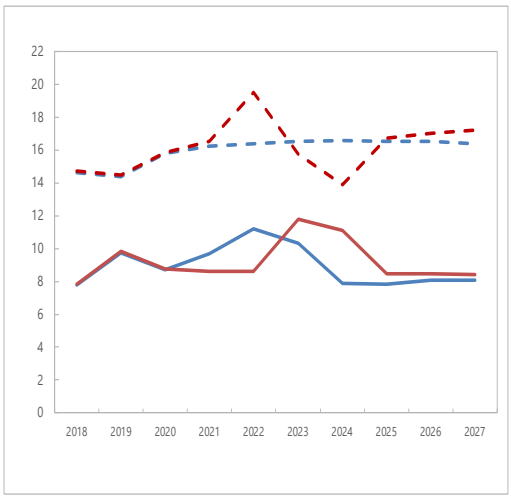
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/

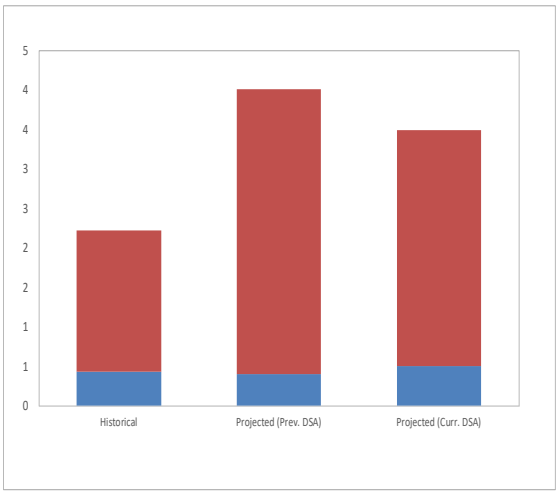


1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Contribution to Real GDP growth
(percent, 5-year average)



Gov. Invest. - Prev. DSA
Gov. Invest. - Curr. DSA
Priv. Invest. - Prev. DSA
Priv. Invest. - Curr. DSA

Contribution of other factors
Contribution of government capital

Sources: Country authorities; and staff estimates and projections.

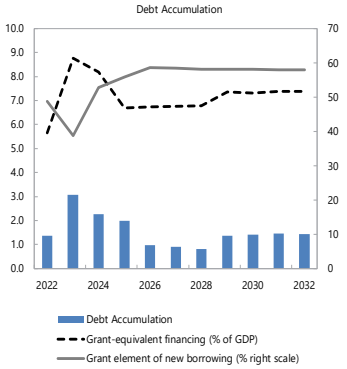
The significant difference between 'Prev. DSA' and 'Çurr. DSA' lines in the 'Public and Private Investment Rates' chart is predominantly due to a calculation error in the previous DSA. Specifically, foreign financed public investment was erroneously included in private investment rather than public investment, hence artificially inflating the former and deflating the latter. The sum of private and public investment in the two DSA versions is broadly similar.

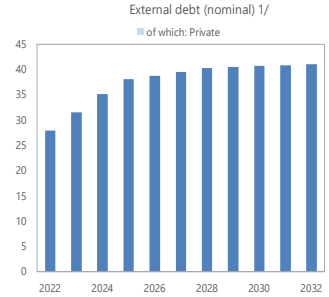
Table 1. Comoros: External Debt Sustainability Framework

(In percent of GDP, unless otherwise indicated)

	Actual			Projections												Average 8/	
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2042	Historical	Projections
External debt (nominal) 1/	**20.3**	**23.2**	**24.7**	**27.9**	**31.5**	**35.1**	**38.0**	**38.7**	**39.5**	**40.3**	**40.5**	**40.7**	**40.8**	**41.0**	**43.5**	**17.5**	**37.6**
of which: public and publicly guaranteed (PPG)	**20.3**	**23.2**	**24.7**	**27.9**	**31.5**	**35.1**	**38.0**	**38.7**	**39.5**	**40.3**	**40.5**	**40.7**	**40.8**	**41.0**	**43.5**	**17.5**	**37.6**
Change in external debt	4.5	2.9	1.4	3.2	3.7	3.6	2.9	0.7	0.8	0.8	0.2	0.2	0.2	0.1	1.6		
Identified net debt-creating flows	**3.0**	**1.1**	**-1.9**	**3.6**	**5.0**	**4.2**	**4.1**	**4.2**	**3.2**	**2.5**	**2.6**	**2.9**	**2.9**	**2.8**	**2.6**	**1.6**	**3.5**
Non-interest current account deficit	**3.3**	**1.7**	**-0.6**	**4.3**	**6.8**	**5.9**	**5.7**	**6.0**	**4.9**	**4.3**	**4.3**	**4.6**	**4.6**	**4.6**	**4.4**	**2.5**	**5.1**
Deficit in balance of goods and services	17.9	22.7	21.0	23.2	25.1	23.7	21.8	21.1	20.1	19.5	19.5	19.9	19.9	19.9	19.7	**19.0**	**21.3**
Exports	11.9	5.5	11.5	12.4	11.7	11.8	12.0	12.3	12.6	13.0	13.5	13.4	13.7	13.9	15.6		
Imports	29.8	28.2	32.5	35.5	36.8	35.5	33.8	33.4	32.7	32.5	33.0	33.3	33.6	33.8	35.3		
Net current transfers (negative = inflow)	-14.0	-20.2	-20.8	-18.1	-17.7	-17.1	-15.5	-14.7	-14.7	-14.8	-14.8	-14.9	-15.0	-15.0	-15.0	**-16.1**	**-15.7**
of which: official	-1.8	-3.4	-1.8	-1.2	-2.2	-2.2	-1.9	-1.9	-1.9	-1.9	-1.9	-1.9	-1.9	-1.9	-1.7		
Other current account flows (negative = net inflow)	-0.7	-0.7	-0.7	-0.8	-0.6	-0.6	-0.6	-0.5	-0.5	-0.4	-0.4	-0.4	-0.3	-0.3	-0.4	**-0.5**	**-0.5**
Net FDI (negative = inflow)	**-0.4**	**-0.3**	**-0.3**	**-0.4**	**-1.5**	**-1.1**	**-0.6**	**-0.6**	**-0.6**	**-0.6**	**-0.6**	**-0.6**	**-0.6**	**-0.6**	**-0.6**	**-0.5**	**-0.7**
Endogenous debt dynamics 2/	**0.1**	**-0.3**	**-1.0**	**-0.3**	**-0.3**	**-0.6**	**-0.9**	**-1.1**	**-1.2**	**-1.2**	**-1.1**	**-1.1**	**-1.2**	**-1.1**	**-1.2**		
Contribution from nominal interest rate	0.2	0.2	0.3	0.3	0.5	0.5	0.5	0.4	0.4	0.4	0.4	0.4	0.3	0.3	0.3		
Contribution from real GDP growth	-0.3	0.0	-0.5	-0.6	-0.8	-1.1	-1.4	-1.6	-1.6	-1.6	-1.5	-1.5	-1.5	-1.5	-1.5		
Contribution from price and exchange rate changes	0.2	-0.5	-0.8	...	...	...	...	...	...	...	...	...	...	...	...		
Residual 3/	**1.5**	**1.8**	**3.3**	**-0.3**	**-1.4**	**-0.7**	**-1.2**	**-3.5**	**-2.4**	**-1.7**	**-2.4**	**-2.8**	**-2.7**	**-2.7**	**-1.0**	**-1.8**	**-2.0**
of which: exceptional financing	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators																	
PV of PPG external debt-to-GDP ratio	**...**	**...**	**16.2**	**18.3**	**19.6**	**20.9**	**22.0**	**22.3**	**22.6**	**23.0**	**23.0**	**23.0**	**23.1**	**23.2**	**23.5**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**140.7**	**148.1**	**167.6**	**176.6**	**183.7**	**181.6**	**179.8**	**177.2**	**170.9**	**171.5**	**169.3**	**166.9**	**150.7**		
PPG debt service-to-exports ratio	**5.4**	**11.8**	**6.1**	**6.3**	**12.2**	**14.0**	**16.8**	**15.7**	**15.5**	**14.8**	**8.4**	**7.8**	**7.6**	**7.7**	**7.8**		
PPG debt service-to-revenue ratio	**6.6**	**5.5**	**6.9**	**8.1**	**15.1**	**17.1**	**20.1**	**18.9**	**18.8**	**18.2**	**10.5**	**9.5**	**9.3**	**9.4**	**9.0**		
Gross external financing need (Million of U.S. dollars)	42.0	25.1	-2.5	57.6	91.1	91.6	103.9	110.3	97.5	89.1	80.6	89.1	95.5	100.2	175.5		
Key macroeconomic assumptions																	
Real GDP growth (in percent)	1.8	-0.2	2.1	2.4	3.0	3.6	4.1	4.3	4.3	4.2	3.8	3.8	3.8	3.8	3.8	**2.5**	**3.7**
GDP deflator in US dollar terms (change in percent)	-1.0	2.7	3.4	-6.3	6.2	0.8	-0.3	-0.9	-1.6	-2.3	2.0	2.0	2.0	2.0	2.0	**0.0**	**0.3**
Effective interest rate (percent) 4/	1.2	1.0	1.2	1.1	1.8	1.6	1.4	1.2	1.1	1.1	1.0	1.0	0.8	0.8	0.8	**0.7**	**1.2**
Growth of exports of G&S (US dollar terms, in percent)	-4.9	-52.1	119.2	3.1	3.2	5.9	5.3	5.6	5.1	5.1	9.9	5.8	7.7	7.7	1.6	**11.1**	**5.8**
Growth of imports of G&S (US dollar terms, in percent)	-0.7	-2.9	21.6	5.0	13.3	0.7	-1.2	2.2	0.5	1.2	7.3	7.0	6.9	6.5	5.7	**3.4**	**4.5**
Grant element of new public sector borrowing (in percent)	...	...	...	48.8	38.8	52.8	55.8	58.6	58.4	58.1	58.2	58.1	58.0	58.0	57.9	**...**	**54.9**
Government revenues (excluding grants, in percent of GDP)	9.6	12.0	10.3	9.6	9.5	9.7	10.0	10.2	10.4	10.6	10.8	11.0	11.2	11.4	13.5	**14.2**	**10.4**
Aid flows (in Million of US dollars) 5/	73.7	75.8	86.8	78.0	134.3	134.8	109.6	119.5	122.1	122.7	145.7	152.5	163.5	173.0	366.6		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	5.6	8.8	8.2	6.7	6.7	6.8	6.8	7.4	7.3	7.4	7.4	8.4	**...**	**7.2**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	80.5	58.4	74.1	74.7	85.3	85.9	87.2	84.6	85.1	84.8	84.8	80.6	**...**	**80.5**
Nominal GDP (Million of US dollars)	1,188	1,218	1,285	1,233	1,349	1,409	1,463	1,512	1,552	1,580	1,673	1,772	1,876	1,986	3,518		
Nominal dollar GDP growth	0.8	2.5	5.5	-4.0	9.4	4.4	3.8	3.4	2.6	1.8	5.9	5.9	5.9	5.9	5.9	**2.6**	**4.1**
Memorandum items:																	
PV of external debt 7/	...	...	16.2	18.3	19.6	20.9	22.0	22.3	22.6	23.0	23.0	23.0	23.1	23.2	23.5		
In percent of exports	...	...	140.7	148.1	167.6	176.6	183.7	181.6	179.8	177.2	170.9	171.5	169.3	166.9	150.7		
Total external debt service-to-exports ratio	5.4	11.8	6.1	6.3	12.2	14.0	16.8	15.7	15.5	14.8	8.4	7.8	7.6	7.7	7.8		
PV of PPG external debt (in Million of US dollars)			208.3	226.0	263.9	294.4	322.5	336.7	350.4	362.9	384.5	408.1	433.9	460.7	825.7		
(PVt-PVt-1)/GDPt-1 (in percent)				1.4	3.1	2.3	2.0	1.0	0.9	0.8	1.4	1.4	1.5	1.4	2.0		
Non-interest current account deficit that stabilizes debt ratio	-1.2	-1.2	-2.0	1.0	3.1	2.4	2.8	5.2	4.2	3.5	4.1	4.4	4.5	4.4	2.8		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

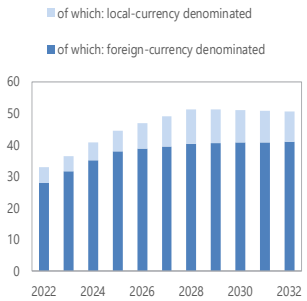
Table 2. Comoros: Public Sector Debt Sustainability Framework

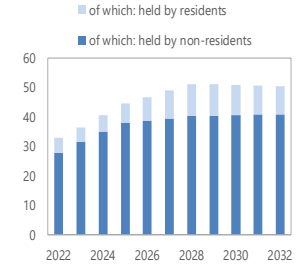
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2032	2042	Historical	Projections
Public sector debt 1/	**23.0**	**26.1**	**29.8**	**33.0**	**36.4**	**40.7**	**44.5**	**46.7**	**49.0**	**50.5**	**47.9**	**19.4**	**45.9**
of which: external debt	**20.3**	**23.2**	**24.7**	**27.9**	**31.5**	**35.1**	**38.0**	**38.7**	**39.5**	**41.0**	**43.5**	**17.5**	**37.6**
Change in public sector debt	4.8	3.2	3.7	3.2	3.4	4.3	3.8	2.3	2.3	-0.2	-0.4		
Identified debt-creating flows	**3.6**	**-0.3**	**1.6**	**3.8**	**5.8**	**4.6**	**2.6**	**1.1**	**0.6**	**-0.2**	**0.4**	**-0.9**	**1.6**
Primary deficit	**3.6**	**0.0**	**2.5**	**3.4**	**6.0**	**5.2**	**4.0**	**2.7**	**2.5**	**2.1**	**1.8**	**-0.5**	**3.1**
Revenue and grants	15.8	18.3	17.0	13.9	14.3	14.7	13.8	15.3	15.6	16.9	19.1	**17.9**	**15.5**
of which: grants	6.2	6.2	6.8	4.3	4.8	5.0	3.8	5.1	5.2	5.6	5.6		
Primary (noninterest) expenditure	19.5	18.2	19.5	17.3	20.3	19.9	17.9	18.0	18.1	19.0	20.9	**17.4**	**18.6**
Automatic debt dynamics	**0.0**	**-0.2**	**-0.9**	**-1.0**	**-1.1**	**-1.4**	**-1.9**	**-2.1**	**-2.2**	**-2.2**	**-1.4**		
Contribution from interest rate/growth differential	-0.1	0.2	-0.7	-1.0	-1.1	-1.4	-1.9	-2.1	-2.2	-2.2	-1.4		
of which: contribution from average real interest rate	0.3	0.1	-0.2	-0.3	-0.1	-0.2	-0.3	-0.3	-0.3	-0.4	0.4		
of which: contribution from real GDP growth	-0.3	0.0	-0.5	-0.7	-1.0	-1.3	-1.6	-1.8	-1.9	-1.9	-1.8		
Contribution from real exchange rate depreciation	0.1	-0.4	-0.2	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**1.4**	**0.9**	**0.8**	**0.5**	**0.5**	**0.2**	**0.0**	**0.0**	**0.0**	**0.4**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	-0.4	-0.6	-0.3	-0.3	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	1.8	1.4	1.1	0.8	0.5	0.2	0.0	0.0		
Residual	**1.2**	**3.4**	**2.1**	**-0.6**	**-2.4**	**-0.2**	**1.2**	**1.2**	**1.7**	**-0.1**	**-0.8**	**1.2**	**0.2**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**21.4**	**23.4**	**24.4**	**26.5**	**28.5**	**30.3**	**32.1**	**32.7**	**27.9**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**125.4**	**168.2**	**170.9**	**180.3**	**206.0**	**197.9**	**205.6**	**192.9**	**146.3**		
Debt service-to-revenue and grants ratio 3/	**4.6**	**4.1**	**5.5**	**7.1**	**14.5**	**16.0**	**26.4**	**30.3**	**40.8**	**48.2**	**32.2**		
Gross financing need 4/	4.4	0.7	3.4	4.8	8.6	7.7	7.9	7.3	8.9	10.2	7.9		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	1.8	-0.2	2.1	2.4	3.0	3.6	4.1	4.3	4.3	3.8	3.8	**2.5**	**3.7**
Average nominal interest rate on external debt (in percent)	1.2	1.0	1.2	1.1	1.8	1.6	1.4	1.2	1.1	0.8	0.8	**0.7**	**1.2**
Average real interest rate on domestic debt (in percent)	-1.6	1.7	3.4	-3.5	-2.8	0.0	-0.6	-0.2	0.2	0.5	0.6	**0.6**	**-0.4**
Real exchange rate depreciation (in percent, + indicates depreciation)	0.3	-1.9	-0.9	...	...	...	...	...	...	...	...	**1.5**	**...**
Inflation rate (GDP deflator, in percent)	4.5	0.8	-0.3	5.2	4.4	1.5	2.3	2.1	2.0	2.0	2.0	**1.4**	**2.5**
Growth of real primary spending (deflated by GDP deflator, in percent)	3.3	-6.6	9.4	-9.1	20.6	1.4	-6.3	5.3	5.0	4.6	4.4	**7.2**	**3.7**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-1.2	-3.2	-1.2	0.3	2.6	0.8	0.2	0.5	0.3	2.3	2.2	**-1.9**	**1.3**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, central bank, government-guaranteed debt . Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Comoros: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2022-2032

(In percent)

	Projections 1/										
	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
PV of debt-to GDP ratio											
Baseline	18	20	21	22	22	23	23	23	23	23	23
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	18	19	19	19	18	17	16	16	16	15	15
B. Bound Tests											
B1. Real GDP growth	18	20	22	24	24	24	25	25	25	25	25
B2. Primary balance	18	21	24	26	27	27	28	28	28	28	28
B3. Exports	18	23	29	30	30	31	31	31	31	31	31
B4. Other flows 3/	18	22	26	27	28	28	29	28	28	28	28
B5. Depreciation	18	24	22	23	24	24	24	25	25	25	25
B6. Combination of B1-B5	18	25	27	29	29	29	30	30	30	30	30
C. Tailored Tests											
C1. Combined contingent liabilities	18	23	26	28	29	29	30	30	30	30	30
C2. Natural disaster	18	23	25	27	28	29	29	30	30	30	30
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	148	168	177	**184**	**182**	180	177	171	171	169	167
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	148	164	164	160	146	135	125	119	116	112	107
B. Bound Tests											
B1. Real GDP growth	148	168	177	**184**	**182**	180	177	171	171	169	167
B2. Primary balance	148	179	**202**	**217**	**219**	**219**	**217**	**210**	**210**	**207**	**203**
B3. Exports	148	**331**	**693**	**713**	**704**	**697**	**687**	**660**	**660**	**648**	**636**
B4. Other flows 3/	148	**192**	**222**	**228**	**226**	**223**	**220**	**212**	**211**	**208**	**204**
B5. Depreciation	148	168	150	157	155	154	152	147	148	146	145
B6. Combination of B1-B5	148	**254**	**214**	**357**	**352**	**349**	**344**	**331**	**331**	**325**	**320**
C. Tailored Tests											
C1. Combined contingent liabilities	148	**195**	**217**	**231**	**232**	**233**	**231**	**222**	**223**	**219**	**215**
C2. Natural disaster	148	**197**	**218**	**231**	**233**	**233**	**232**	**224**	**225**	**222**	**218**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	6	12	14	**17**	**16**	**16**	15	8	8	8	8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	6	13	15	**18**	**17**	**16**	**15**	8	8	8	8
B. Bound Tests											
B1. Real GDP growth	6	12	14	**17**	**16**	**16**	15	8	8	8	8
B2. Primary balance	6	12	14	**17**	**16**	**16**	**16**	9	8	8	8
B3. Exports	6	**21**	**44**	**54**	**51**	**50**	**48**	**28**	**26**	**25**	**25**
B4. Other flows 3/	6	12	15	**18**	**17**	**16**	**16**	9	8	8	8
B5. Depreciation	6	12	14	**16**	**15**	**15**	14	8	7	7	7
B6. Combination of B1-B5	6	**16**	**24**	**28**	**27**	**26**	**25**	15	13	13	13
C. Tailored Tests											
C1. Combined contingent liabilities	6	12	15	**18**	**17**	**17**	**16**	9	9	8	9
C2. Natural disaster	6	13	15	**18**	**17**	**17**	**16**	10	9	9	9
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	8	15	17	**20**	**19**	**19**	**18**	11	10	9	9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	8	16	**18**	**21**	**20**	**20**	**18**	10	9	9	10
B. Bound Tests											
B1. Real GDP growth	8	16	**18**	**22**	**20**	**20**	**19**	11	10	10	10
B2. Primary balance	8	15	17	**21**	**20**	**20**	**19**	11	10	10	10
B3. Exports	8	16	**19**	**23**	**21**	**21**	**21**	12	11	11	11
B4. Other flows 3/	8	15	18	**21**	**20**	**20**	**19**	11	10	10	10
B5. Depreciation	8	**19**	**21**	**24**	**23**	**23**	**22**	12	11	11	11
B6. Combination of B1-B5	8	16	**19**	**23**	**21**	**21**	**21**	12	11	11	11
C. Tailored Tests											
C1. Combined contingent liabilities	8	15	18	**21**	**20**	**20**	**19**	12	11	10	10
C2. Natural disaster	8	15	18	**21**	**20**	**20**	**19**	12	11	10	10
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Comoros: Sensitivity Analysis for Key Indicators of Public Debt, 2022-2032

	Projections 1/										
	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
PV of Debt-to-GDP Ratio											
Baseline	23	24	27	29	30	32	34	34	33	33	33
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	23	20	21	20	20	23	23	23	21	19	20
B. Bound Tests											
B1. Real GDP growth	23	25	29	31	33	35	38	38	38	38	38
B2. Primary balance	23	27	32	33	34	36	38	37	37	36	36
B3. Exports	23	27	33	35	36	38	40	40	39	39	39
B4. Other flows 3/	23	27	32	34	36	38	39	39	39	38	38
B5. Depreciation	23	27	27	28	28	29	30	29	28	27	26
B6. Combination of B1-B5	23	26	28	28	29	30	32	31	31	31	30
C. Tailored Tests											
C1. Combined contingent liabilities	23	33	34	35	36	37	39	39	38	38	38
C2. Natural disaster	23	32	33	34	35	37	39	39	38	38	38
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
TOTAL public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	168	171	180	206	198	206	212	206	202	197	193
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	168	138	141	142	129	140	136	131	118	107	107
B. Bound Tests											
B1. Real GDP growth	168	174	190	220	212	222	231	226	224	221	219
B2. Primary balance	168	192	218	238	223	229	235	228	223	217	213
B3. Exports	168	188	222	251	238	246	253	245	239	233	228
B4. Other flows 3/	168	191	217	245	233	241	247	240	234	228	224
B5. Depreciation	168	198	191	203	188	190	193	181	173	164	157
B6. Combination of B1-B5	168	181	189	202	190	194	199	192	187	183	180
C. Tailored Tests											
C1. Combined contingent liabilities	168	234	229	250	234	240	246	238	233	227	223
C2. Natural disaster	168	223	221	244	230	237	243	236	232	227	223
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	7	14	16	26	30	41	50	54	52	50	48
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	7	15	(5)	27	17	26	53	49	56	46	40
B. Bound Tests											
B1. Real GDP growth	7	15	19	34	38	50	59	63	60	58	56
B2. Primary balance	7	15	33	54	44	50	56	58	54	52	50
B3. Exports	7	14	16	27	31	42	51	55	52	51	49
B4. Other flows 3/	7	14	16	27	31	41	51	55	52	50	49
B5. Depreciation	7	16	20	33	36	46	54	55	52	50	48
B6. Combination of B1-B5	7	14	16	35	34	44	51	54	51	49	48
C. Tailored Tests											
C1. Combined contingent liabilities	7	15	60	52	44	50	56	58	55	52	50
C2. Natural disaster	7	15	52	49	43	50	57	59	56	53	51
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the benchmark.

2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

3/ Includes official and private transfers and FDI.